

1.5 Surface Area Worksheet

1. Setup, but do not evaluate, an integral for the surface area of the surface obtained by rotating $y = \tan(x)$

on $\left[0, \frac{\pi}{3}\right]$ about the...

a) x -axis (setup integral with respect to x and y)

b) y -axis (setup integrals with respect to x and y)

2. Setup, but do not evaluate, an integral for the surface area of the surface obtained by rotating $x = \ln(2y + 1)$ on $0 \leq y \leq 1$ about the...
- a) y -axis (create integrals with respect to x and y).

b) x -axis

3. Find the surface area of the solid generated by revolving the curve $y = \sqrt{x}$, $1 \leq x \leq 4$ about the x -axis. Set up integrals with respect to x and y and then evaluate one of them.

4. Find the surface area of the solid generated by revolving the curve $x = 2\sqrt{1+y}$, $0 \leq y \leq 1$ about the y -axis. Set up integrals with respect to x and y and then evaluate one of them.

5. Consider the curve $y = \sqrt[3]{x}$ on the interval $0 \leq y \leq 2$ and the resulting surface when this curve is rotated about the y -axis. Setup and evaluate, an integral that represents the surface area of this surface.
- a) Setup and evaluate the integral with respect to y .

6. Find the surface area of the solid generated by revolving $y = x^2$ on $[0, 2]$ about the y -axis.

7. Consider the curve $y = \sin(\pi x)$ on the interval $0 \leq x \leq 1$ and the resulting surface when this curve is rotated about the x -axis. Setup, but to not evaluate, an integral that represents the surface area of this surface.

a) Setup the integral with respect to x .

b) Explain why setting up this integral with respect to y would be very difficult.